

Week 1 (DSA)

Exercise 2: E-commerce Platform Search Function

Product.java

```
package com.cognizant.search;

public class Product {

    private int productId;
    private String productName;
    private String category;

    public Product(int productId, String productName, String category) {
        this.productId = productId;
        this.productName = productName;
        this.category = category;
    }

    public int getProductId() {
        return productId;
    }

    public String getProductName() {
        return productName;
    }

    public String getCategory() {
        return category;
    }

    @Override
    public String toString() {
        return "Product ID: " + productId +
            ", Name: " + productName +
            ", Category: " + category;
    }
}
```

SearchOperation.java

```
package com.cognizant.search;

public class SearchOperations {

    // Linear Search
    public static Product linearSearch(Product[] products, int id) {

        for (Product product : products) {
            if (product.getProductId() == id) {
                return product;
            }
        }
    }
}
```

```

        return null;
    }

    // Binary Search
    public static Product binarySearch(Product[] products, int id) {

        int left = 0;
        int right = products.length - 1;

        while (left <= right) {

            int mid = (left + right) / 2;

            if (products[mid].getProductId() == id) {
                return products[mid];
            }

            else if (products[mid].getProductId() < id) {
                left = mid + 1;
            }

            else {
                right = mid - 1;
            }
        }

        return null;
    }
}

```

SearchTest.java

```

package com.cognizant.search;

public class SearchTest {

    public static void main(String[] args) {

        Product[] products = {

            new Product(101, "Laptop", "Electronics"),
            new Product(102, "Shoes", "Fashion"),
            new Product(103, "Phone", "Electronics"),
            new Product(104, "Watch", "Accessories"),
            new Product(105, "Book", "Education")

        };

        System.out.println("Linear Search:");

        Product result1 = SearchOperations.linearSearch(products, 104);

        if(result1 != null)
            System.out.println(result1);
        else
    }
}

```

```
        System.out.println("Product Not Found");

    }

    System.out.println("\nBinary Search:");

    Product result2 = SearchOperations.binarySearch(products, 104);

    if(result2 != null)
        System.out.println(result2);
    else
        System.out.println("Product Not Found");
}
}
```

Output



```
Run SearchTest x
C:\Program Files\Java\jdk-21\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ...
Linear Search:
Product ID: 104, Name: Watch, Category: Accessories

Binary Search:
Product ID: 104, Name: Watch, Category: Accessories

Process finished with exit code 0
```